

SIMATS ENGINEERING
ASSESSMENT TOOL 3 - LOW (Coding Challenge)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications. (BL3, BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks: 50

ANNEXURE A

CODING CHALLENGE

Code of Conduct:

*I S.K.Deepa Arathi (192571020) that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Rubrics for Evaluation:

Criteria	Problem Understanding	Algorithm	Code Implementation	Competitive Performance	Test Case Handling	Total (30%)	Total (100)
Weightage	20%	25%	25%	20%	15%		
Q1- Q2							
Q3-Q4							
Q5-Q6							
Q7-Q8							
Q9-Q10							
Total							

Q1. The following Java program is intended to print all elements of an array. However, it throws an error during execution. Identify and correct the mistake in the loop condition.

```
public class Q1_ArrayIndex {
    public static void main(String[] args) {
        int arr[] = {10, 20, 30};
        for(int i = 0; i < arr.length; i++){
            System.out.println(arr[i]);
        }
    }
}
```

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the text "SIMATS Online Programming Lab", and the user's name "S K DEEPA ARATHI" with a profile icon. Below the header, there is a sidebar on the left with a list of questions (Q5 to Q10) and their respective marks. The main area is titled "Your Code" and contains a Java program. The program is intended to print the elements of an array, but it has a mistake in the loop condition. The code is as follows:

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int[] arr = {10, 20, 30};
8
9         for (int i = 0; i < arr.length; i++) {
10             System.out.println(arr[i]);
11         }
12     }
13 }
```

The output of the program is shown on the right, displaying the elements of the array: 10, 20, and 30. Below the output, there is a section for "Visible Test Cases" with three cases: Test Case 1, Test Case 2, and Test Case 3. At the bottom of the main area, there are buttons for "Run" and "Save".

Q2. A program attempts to print the length of a string but results in a runtime exception. Analyze the issue and modify the code to handle it properly.

```
public class Q2_NullPointer {
    public static void main(String[] args) {
        String str = null;
        if(str != null)
            System.out.println(str.length());
        else
            System.out.println("String is null");
    }
}
```

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Q4 While Loop Correction
5 marks

Q5 Reverse String Logic
5 marks

Q6 Array Sum Calculation
5 marks

Q7 String Character Access
5 marks

Q8 Case Insensitive Comparison
5 marks

Q9 Array Declaration
5 marks

Q10 Switch Statement
5 marks

10/10 answered

```

System.out.println("String is null");
}
}

```

Your Code

```

1 import java.util.Scanner;
2 public class Q2_NullPointer {
3     public static void main(String[] args) {
4         String str = null;
5
6         if (str != null) {
7             System.out.println(str.length());
8         } else {
9             System.out.println("String is null");
10        }
11    }
12 }

```

Input

stdin input

Output

String is null

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

Run **Save**

Q3. A developer compares two strings using the == operator, which leads to incorrect output. Identify the issue and correct the comparison method.

```

public class Q3_StringCompare {
    public static void main(String[] args) {
        String s1 = "Java";
        String s2 = new String("Java");

        if(s1.equals(s2)){
            System.out.println("Equal");
        }
    }
}

```

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Q5 Reverse String Logic
5 marks

Q6 Array Sum Calculation
5 marks

Q7 String Character Access
5 marks

Q8 Case Insensitive Comparison
5 marks

Q9 Array Declaration
5 marks

Q10 Switch Statement
5 marks

10/10 answered

```

1 import java .util.Scanner;
2 public class Q3_StringCompare {
3     public static void main(String[] args) {
4         String s1 = "Java";
5         String s2 = new String("Java");
6
7         if (s1.equals(s2)) {
8             System.out.println("Equal");
9         }
10    }
11 }

```

Input

stdin input

Output

Equal

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

Run **Save**

Q4. The given program uses a loop to print numbers from 0 to 4, but it runs infinitely. Identify the cause and fix the loop.

```
public class Q4_Loop {  
    public static void main(String[] args) {  
        int i = 0;  
        while(i < 5){  
            System.out.println(i);  
            i++;  
        }  
    }  
}
```



The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user "S K DEEPA ARATHI" with a dropdown arrow. Below the header, there is a sidebar with a list of questions (Q5 to Q10) and their marks. The main area is titled "Your Code" and contains the following Java code:

```
1 import java.util.Scanner;  
2 public class Q4_Loop {  
3     public static void main(String[] args) {  
4         int i = 0;  
5         while (i < 5) {  
6             System.out.println(i);  
7             i++;  
8         }  
9     }  
10 }
```

The "Input" field is empty, and the "Output" field shows the numbers 0, 1, 2, 3, 4, and then the program continues to run. The "Visible Test Cases" section shows three test cases, each with a status indicator.

Q5. A program is written to reverse a string. Verify whether the logic is correct and modify it if necessary.

```
public class Q5_Reverse {  
    public static void main(String[] args) {  
        String str = "Hello";  
        String rev = "";  
  
        for(int i = 0; i < str.length(); i++){  
            rev = str.charAt(i) + rev;  
        }  
        System.out.println(rev);  
    }  
}
```

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Q5
Reverse String Logic
5 marks

Q6
Array Sum Calculation
5 marks

Q7
String Character Access
5 marks

Q8
Case Insensitive Comparison
5 marks

Q9
Array Declaration
5 marks

Q10
Switch Statement
5 marks

10/10 answered

Your Code

```

1 import java.util.Scanner;
2 public class Q5_Reverse {
3     public static void main(String[] args) {
4         String str = "Hello";
5         String rev = "";
6         for (int i = str.length() - 1; i >= 0; i--) {
7             rev += str.charAt(i);
8         }
9         System.out.println(rev);
10    }
11 }

```

Run Save

Input

stdin input

Output

olleH

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

Q6. Identify the problem and fix it.

```

public class Q6_ArraySum {
    public static void main(String[] args) {
        int arr[] = {1,2,3,4};
        int sum = 0;
        for(int i=0;i<arr.length;i++){
            sum += arr[i];
        }
        System.out.println(sum);
    }
}

```

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Q5
Reverse String Logic
5 marks

Q6
Array Sum Calculation
5 marks

Q7
String Character Access
5 marks

Q8
Case Insensitive Comparison
5 marks

Q9
Array Declaration
5 marks

Q10
Switch Statement
5 marks

10/10 answered

Your Code

```

1 import java .util.Scanner;
2 public class Q6_ArraySum {
3     public static void main(String[] args) {
4         int[] arr = {1, 2, 3, 4};
5         int sum = 0;
6
7         for (int i = 0; i < arr.length; i++) {
8             sum += arr[i];
9         }
10
11         System.out.println(sum);
12    }
13 }

```

Run Save

Input

stdin input

Output

10

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

Q7. A program tries to access a character from a string using an invalid index, causing an exception. Identify and correct the issue.

```
public class Q7_StringIndex {  
    public static void main(String[] args) {  
        String str = "Java";  
        System.out.println(str.charAt(3));  
    }  
}
```

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Q7 String Character Access (5 marks) is selected in the sidebar. The code in the editor is:

```
1 import java.util.Scanner;  
2 public class Q7_StringIndex {  
3     public static void main(String[] args) {  
4         String str = "Java";  
5         System.out.println(str.charAt(0));  
6     }  
7 }
```

The "Input" field contains "stdin input". The "Output" field shows "J". The "Visible Test Cases" section lists "Test Case 1", "Test Case 2", and "Test Case 3", each with a status indicator.

Q8. A string comparison fails due to case sensitivity. Modify the program to make the comparison case-insensitive.

```
public class Q8_CaseInsensitive {  
    public static void main(String[] args) {  
        String str = "java";  
        if(str.equalsIgnoreCase("JAVA")){  
            System.out.println("Match");  
        }  
    }  
}
```

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5 marks
Q4
 While Loop Correction
 5 marks
 ✓

5 marks
Q5
 Reverse String Logic
 5 marks
 ✓

5 marks
Q6
 Array Sum Calculation
 5 marks
 ✓

5 marks
Q7
 String Character Access
 5 marks
 ✓

5 marks
Q8
 Case Insensitive Comparison
 5 marks
 ✓

5 marks
Q9
 Array Declaration
 5 marks
 ✓

5 marks
Q10
 Switch Statement
 5 marks
 ✓

```

1  }
2  }
    
```

Your Code

```

1  import java.util.Scanner;
2  public class Q9_CaseInsensitive {
3      public static void main(String[] args) {
4          String str = "java";
5          if (str.equalsIgnoreCase("JAVA")) {
6              System.out.println("Match");
7          }
8      }
9  }
    
```

Input
 stdin input

Output
 Match

Visible Test Cases

Test Case 1	●
Test Case 2	●
Test Case 3	●

Run Save

Q9. The program contains an error in array declaration and initialization. Identify the syntax issue and correct it.

import java.util.Arrays;

```

public class Q9_ArrayDeclaration {
    public static void main(String[] args) {
        int arr[] = {1,2,3};
        System.out.println(Arrays.toString(arr));
    }
}
    
```

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5 marks
Q4
 While Loop Correction
 5 marks
 ✓

5 marks
Q5
 Reverse String Logic
 5 marks
 ✓

5 marks
Q6
 Array Sum Calculation
 5 marks
 ✓

5 marks
Q7
 String Character Access
 5 marks
 ✓

5 marks
Q8
 Case Insensitive Comparison
 5 marks
 ✓

5 marks
Q9
 Array Declaration
 5 marks
 ✓

5 marks
Q10
 Switch Statement
 5 marks
 ✓

```

1  }
2  }
    
```

Your Code

```

1  import java.util.Arrays;
2  public class Q9_ArrayDeclaration {
3      public static void main(String[] args) {
4          int[] arr = {1, 2, 3};
5          System.out.println(Arrays.toString(arr));
6      }
7  }
    
```

Input
 stdin input

Output
 [1, 2, 3]

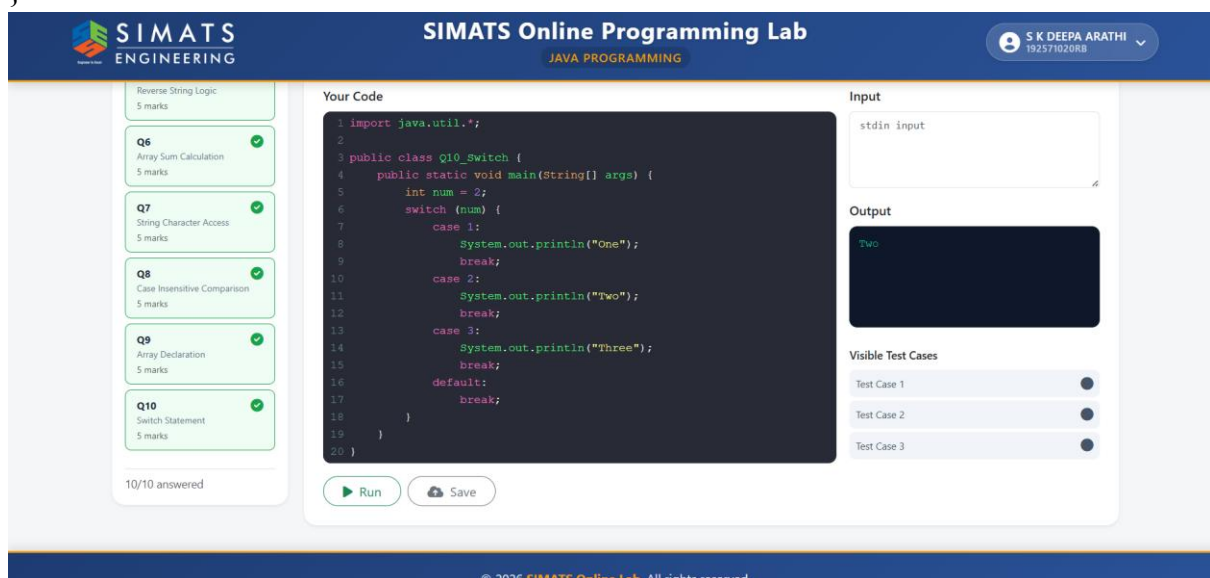
Visible Test Cases

Test Case 1	●
Test Case 2	●
Test Case 3	●

Run Save

Q10. A switch statement prints multiple outputs due to missing statements. Identify the problem and correct the code.

```
public class Q10_Switch {
    public static void main(String[] args) {
        int num = 2;
        switch(num){
            case 1: System.out.println("One"); break;
            case 2: System.out.println("Two"); break;
            case 3: System.out.println("Three"); break;
        }
    }
}
```



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach

Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases